

Shruti Jagdale

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WORK EXPERIENCE

ML Engineer, Agentic Systems | Zeblok Computational | Stony Brook, NY May 2025 - Jan 2026

- Orchestrated multimodal ETL and AI pipeline (text, image) using **LLaMA-70B** via vLLM, LangChain, embedding models (MiniLM, CLIP) for semantic search, **ChromaDB** vector database
- Built Streamlit chatbot UI for RAG pipeline with MCP tool-calling, vector search, and OCR extraction
- Optimized prompt context chunking for the inference pipeline, cutting 50% irrelevant QA outputs with monitoring
- Built ML tooling with SQL-based chat memory, CRUD logic, and embedding storage to enable LLM evaluation and **CoT** reasoning, reducing retrieval time by 20%
- Streamlined **FastAPI** REST APIs in microservice integration for agentic systems, reducing latency by 25%
- Developed recursive agent framework enabling LLaMA-70B to autonomously invoke tools during LLM inference

ML Research Engineer | L3Cube Research Lab | Pune, India Jun 2023 - Jun 2024

- Fine-tuned open-source transformer (Gemma-2B) for Marathi via SFT (PEFT/LoRA) using HuggingFace on GPU
- Increased AI4Bharat benchmark accuracy by 65% through manual QA of code-switched LLMs
- Identified gaps between conventional benchmarks and manual evaluation for low-resource LLM finetuning
- Authored research outcomes in peer-reviewed venues including IEEE, ACL, and Springer
- Trained multilingual embeddings with **Hing-BERT**, Hing-FastText, and Hing-GPT for social media NLP

Data Scientist | FeyNN Labs | Assam, India Mar 2023 - Jul 2023

- Applied clustering (K-Means), feature engineering, data visualization on 10K+ consumer records to model EV adoption
- Improved the baseline accuracy by 59% using ensemble methods (XGBoost, Random Forest); findings informed EV behavior analysis and infrastructure planning decisions
- Benchmarked SVM, KNN, and Logistic Regression for predictive maintenance on sensor data; applied anomaly detection and classification on historical records

EDUCATION

Stony Brook University | Stony Brook, New York Aug 2024 - May 2026

Master of Science in Data Science, GPA: 3.78/4.00

Pune Institute of Computer Technology | Pune, India Aug 2020 - Jun 2024

Bachelor of Engineering, GPA: 3.62/4.00

PROJECTS

Scalable Retrieval System Benchmarking for RAG | Big Data, *Vector Search, Indexing, Metrics* Sep 2025 - Nov 2025

- Built a **RAG** system with dense embeddings on large datasets and implemented FAISS and Spark-based distributed indexing to run large-scale indexing experiments (10k–50k chunks) under CPU-only constraints
- Analyzed and visualized retrieval system performance (p50/p95 latency, throughput), evaluating RAG vs non-RAG tradeoffs and contributing to scalable system design insights

Deep Learning for Order Book Prediction | *Quantitative Finance, BiGRU* Feb 2025 - May 2025

- Designed a High-Frequency Trading workflow on real-time Binance BTC–USDT order book data
- Engineered microstructure features and trained BiGRU sequence algorithms to predict buy pressure (F1 87%) and estimate trading costs (spread, slippage)

Atlas: Full-Stack Agentic Travel Assistant | LangChain, Cursor, Claude Code, Docker Jun 2026

- Architected a full-stack agentic AI assistant using LangChain ReAct agent, MCP tool-calling for multistep reasoning workflows, and Groq LLaMA 3.3 70B in Docker for real-time weather and news recommendations

Multilingual Named Entity Recognition | *HingBERT, Sequence Models, Social Media text* Jun 2023 - Feb 2024

- Refined and validated a BiLSTM-CRF predictor with HingBERT encodings for Hinglish NER, securing 85% F1-score
- Benchmarked against BiLSTM and attention models, showing value of sequence processing in fused-dialect NLP

PUBLICATIONS

Code-Mixed Embeddings for Hate Speech Detection — *Springer Nature, ICCCN 2024* Jun 2025

Adapting LLMs to Low-Resource Settings — *ACL, CHIPSAL 2025* Jan 2025

Code-Mixing in Named Entity Recognition — *IEEE, ICETITE 2024* Feb 2024

TECHNICAL SKILLS

Languages/Tools: Python, R, SQL, Java, C/C++, Streamlit, React, Node.js, Git, Docker, Kubernetes, AWS, Claude Code

ML/AI: NLP, Generative AI, LLMs, Deep Learning, Machine Learning, Prompt Engineering, LLM Evaluation, Model

Deployment, PyTorch, TensorFlow, Scikit-learn, ONNX, Vector Search, Matplotlib, Seaborn, MCP, Sentiment Analysis